



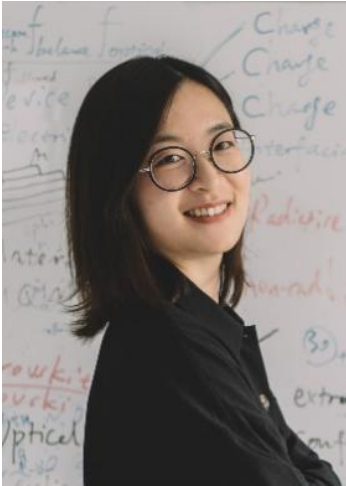
Field and Wave Electromagnetics

电磁场与电磁波

赵保丹

浙江大学光电科学与工程学院

Field and Wave Electromagnetics



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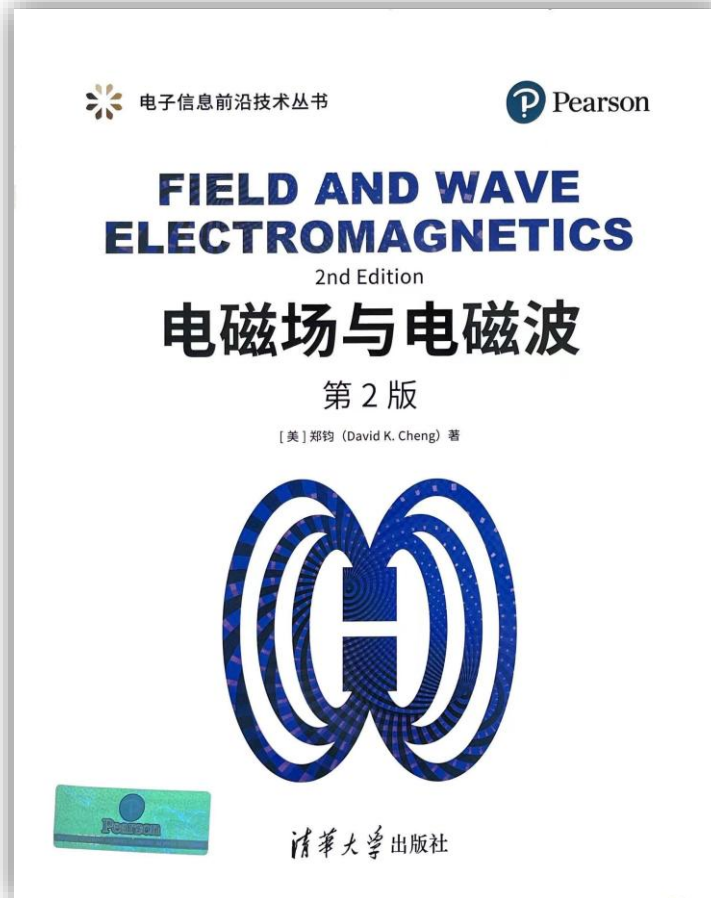
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Field and Wave Electromagnetics

Edited by David K. Cheng

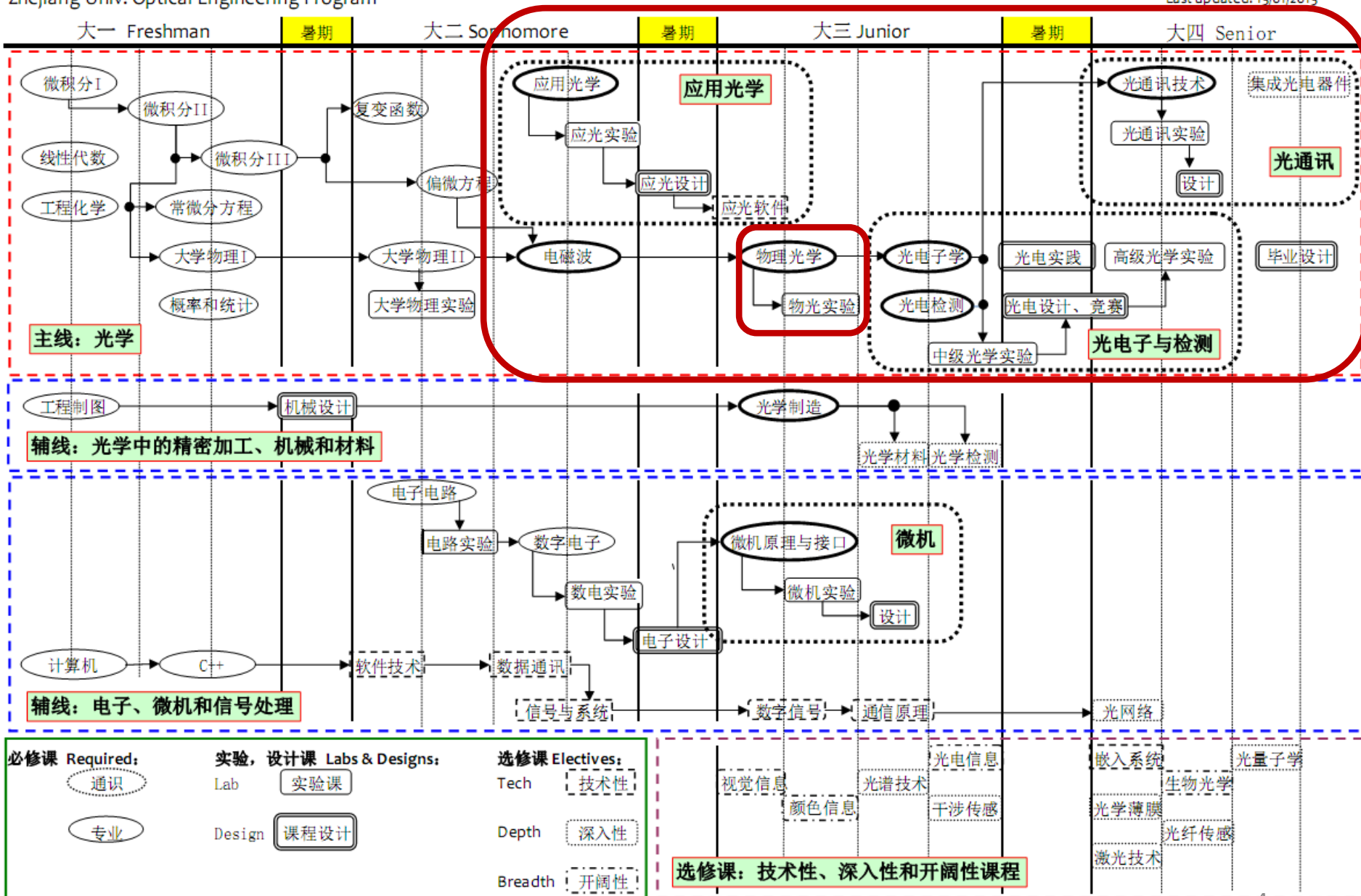


700 pages

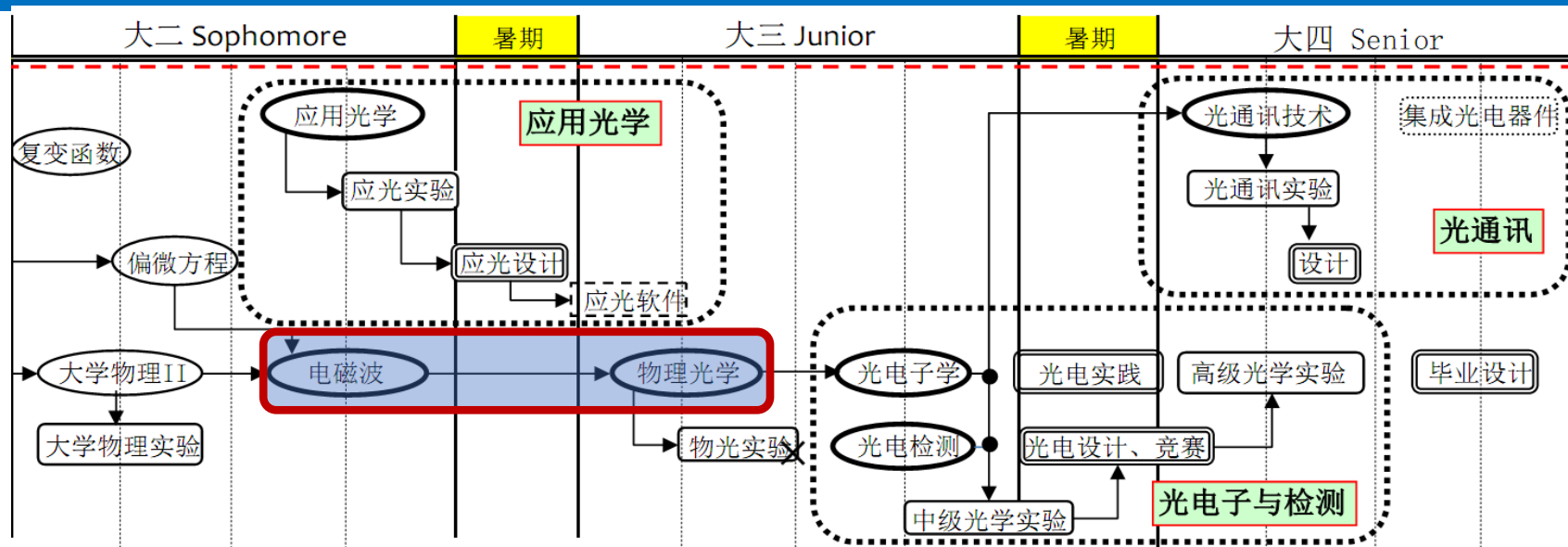
课程流程图
Course Flow Chart

浙江大学光电信息专业
Zhejiang Univ. Optical Engineering Program

Last updated: 13/01/2013



课程地位及特点



最重要的专业基础课程

——承前启后

学分最高的课程

——2.5、4学分

最难的专业基础课程

——内容多/概念多/公式多

光电信
息专业

之一

课程与成绩信息

- 2.5个学分，40个学时，双语
- 春夏{1-16周：2-3学时/周, 不区分单双周}
- 紫金港西2-218(录播)；周二上午第3，4，5节
- 期末考试占总成绩60%，试卷卷面100分（40分小题+60分大题），其中60分大题大部分来源于作业，请大家认真完成作业，可以讨论，但不要抄袭；
- 平时成绩占总成绩40%（即40分），平时成绩由两部分组成，出勤和作业上交程度占20分（交了即满分，不能补交）
作业完成程度占20分（期末考试试卷卷面60分大题原题除以3）

Tips for passing this course

1. Knowing it is the most difficult course.
2. Finishing the homework.
3. Keep pace with the courses; Do not give up.
4. Do not try studying it over in the next year.



瑞斯拜

Chapter Title		课时	
1	Static Electric Fields	3	静电场： (1) 期末<15%，不考大题； (2) 磁矢等难度大尽量简化
2	Solution of Electrostatic Problems		
4	Steady Electric Currents	2	
5	Basic Static Magnetic Fields	2	
7	Time-varying Fields and Maxwell's Equations	5	
9	Plane Electromagnetic Waves	11	电压计算等适当简化，
13	Waveguides and Cavity Resonators	11	重点强化章节，期末有一道推导大题
16	Antennas and Radiating systems	3	
	机动/答疑	2	
		40	

1 总体原则：弱化电路相关(V,A,I,U)，强化Photonics相关(E,H)，突出本课程对光电专业支撑作用

2 删掉“传输线”，增加“天线”；删除“镜像法”（分离变量法保留）

3 课时可以根据情况适当调整，预留2课时为机动/答疑课时；

4 平时成绩40分，（作业上交10分+点名10分，作业完成情况20分）

5 作业完成情况20分学生两种选择（默认选择a）：

（a）作业上交点名20分+作业完成情况(绑定期末作业题)20分

（b）助教随机抽出4个作业大题，每题5分，1小时完成(答疑课时候完成)，得分视为平时成绩。

Maxwell's Equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$



Chapter 1: The Electromagnetic Model

- Conservation of electric charges

The principle of *conservation of electric charge*, like the principle of conservation of momentum, is a fundamental postulate or law of physics. It states that electric charge is conserved; that is, it can neither be created nor be destroyed. This is a law of nature and cannot be derived from other principles or relations. Its truth has never been questioned or doubted in practice.

- Fundamental electromagnetic field quantities

Symbols and Units for Field Quantities	Field Quantity	Symbol	Unit
Electric	Electric field intensity	E	V/m
	Electric flux density (Electric displacement)	D	C/m ²
Magnetic	Magnetic flux density	B	T
	Magnetic field intensity	H	A/m

- Light velocity in vacuum $c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$

Chapter 2: Vector Analysis

1. Vector algebra—addition, subtraction, and multiplication of vectors.
2. Orthogonal coordinate systems—Cartesian, cylindrical, and spherical coordinates.
3. Vector calculus—differentiation and integration of vectors; line, surface, and volume integrals; “del” operator; gradient, divergence, and curl operations.

$$\nabla \equiv \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

Some Useful Vector Identities

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B}$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\nabla(\psi V) = \psi \nabla V + V \nabla \psi$$

$$\nabla \cdot (\psi \mathbf{A}) = \psi \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla \psi$$

$$\nabla \times (\psi \mathbf{A}) = \psi \nabla \times \mathbf{A} + \nabla \psi \times \mathbf{A}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \cdot \nabla V = \nabla^2 V$$

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$\nabla \times \nabla V = 0$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\int_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{s} \quad (\text{Divergence theorem})$$

$$\int_S \nabla \times \mathbf{A} \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \quad (\text{Stokes's theorem})$$

$$\nabla \equiv \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Gradient, Divergence, Curl, and Laplacian Operations

Cartesian Coordinates (x, y, z)

$$\nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \equiv \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \mathbf{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \mathbf{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \mathbf{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical Coordinates (r, ϕ, z)

$$\nabla V = \mathbf{a}_r \frac{\partial V}{\partial r} + \mathbf{a}_\phi \frac{\partial V}{r \partial \phi} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\phi r & \mathbf{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} = \mathbf{a}_r \left(\frac{\partial A_z}{r \partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \mathbf{a}_\phi \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) + \mathbf{a}_z \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right]$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical Coordinates (R, θ, ϕ)

$$\nabla V = \mathbf{a}_R \frac{\partial V}{\partial R} + \mathbf{a}_\theta \frac{1}{R} \frac{\partial V}{\partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\begin{aligned} \nabla \times \mathbf{A} = \frac{1}{R^2 \sin \theta} \begin{vmatrix} \mathbf{a}_R & \mathbf{a}_\theta R & \mathbf{a}_\phi R \sin \theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix} &= \mathbf{a}_R \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &+ \mathbf{a}_\theta \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] \\ &+ \mathbf{a}_\phi \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \end{aligned}$$

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

Chapter 3: Static Electric Fields

Field: A spatial distribution of a scalar or vector quantity which may or may not be a function of time.

1. Fundamental postulates of electrostatic in free space
2. Coulomb's law
3. Gauss's law
4. Conductors in static electric field
5. Dielectrics in static electric field
6. Electric flux density
7. Boundary conditions for electrostatic fields
8. Capacitance and capacitors
9. Electrostatic energy and forces

3.2 Fundamental postulate of electrostatic in free space

◆ Electric field intensity:

$$\vec{E} = \lim_{q \rightarrow 0} \frac{F}{q} \text{ (V/m); Force per unit charge}$$

◆ The two fundamental postulates :

(Electrostatic; free space)

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

ρ : volume density of **charges** (有源的)

ϵ_0 : permittivity of free space

$$\nabla \times \vec{E} = 0$$

→ **Static E fields are irrotational** (无旋的)!

The two equations hold at every spatial point and are referred to as the **differential form** of the postulates.

➤ Take volume integral and apply divergence theorem 体积分化为闭合曲面积分

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \int_V \nabla \cdot \vec{E} dv = \frac{1}{\epsilon_0} \int_V \rho dv \Rightarrow \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

Gauss's Law: the total outward of the electric field intensity **E** over any closed surface is equal to the total charge enclosed by the surface divided by ϵ_0 .

➤ Take line integral and apply Stokes's theorem 把面积分化为环路积分

$$\nabla \times \vec{E} = 0 \Rightarrow \int_S (\nabla \times \vec{E}) \cdot d\vec{s} = 0 \Rightarrow \oint_C \vec{E} \cdot d\vec{l} = 0$$

- Equivalent to the Kirchhoff's voltage law;
- The line integral between two points is irrelevant to the path;
- Static electric field is irrotational (conservative) 无旋的场，也即保守场

These two postulates represent laws of nature.

静电场是无旋（保守）、有源（有散度）的场。（磁场是无源有旋场）。

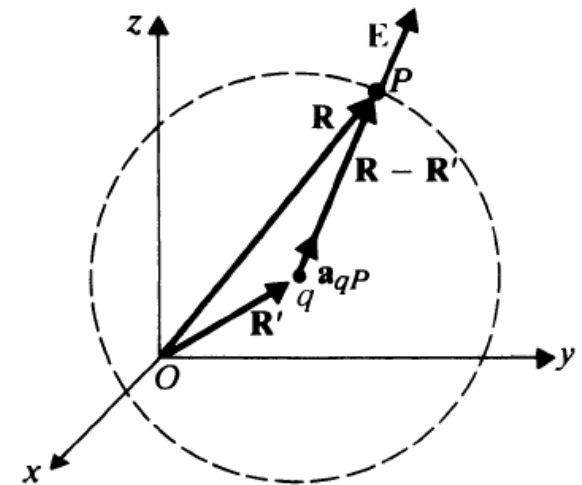
3.3 Coulomb's Law

A point source of charge q at $\mathbf{R}'$ and find the electric field $\mathbf{E}_p$ at $\mathbf{R}$ (at point $\mathbf{P}$) (Apply **Gauss's Law**):

$$\oint_S \vec{E}_P \cdot d\vec{S} = \oint_S (E_P \frac{\vec{R} - \vec{R}'}{|\vec{R} - \vec{R}'|}) \cdot \frac{\vec{R} - \vec{R}'}{|\vec{R} - \vec{R}'|} ds = \frac{q}{\epsilon_0}$$

Unit vector

or $E_P \oint_S ds = E_P (4\pi |\vec{R} - \vec{R}'|^2) = \frac{q}{\epsilon_0}$



Coulomb's Law

$$\vec{E}_p = \frac{q(\vec{R} - \vec{R}')}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|^3}$$



$$\vec{F}_{12} = q_2 \vec{E}_{12} = \frac{q_1 q_2 (\vec{R} - \vec{R}')}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|^3}$$

q_1 at $\mathbf{R}'$ and q_2 at $\mathbf{R}$

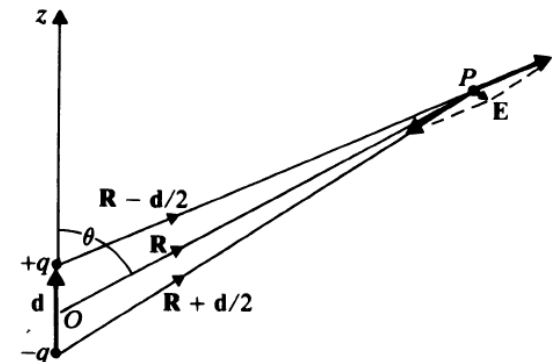
3.3.1 Electric field due to a system of discrete charges

- ◆ **Total field at a point E :** **vector sum** of the fields from all the individual charges

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k (\vec{R} - \vec{R}')}{|\vec{R} - \vec{R}'|^3}$$

注：电单极子（点电荷）的电场强度在数值上与距离平方成反比（且各向同性，与角度 θ 、 ϕ 无关）。

- ◆ **Electric dipole:** a pair of equal and opposite charges $+q$ and $-q$ separated by a small distance d



$$\vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{\vec{R} - \frac{\vec{d}}{2}}{|\vec{R} - \frac{\vec{d}}{2}|^3} - \frac{\vec{R} + \frac{\vec{d}}{2}}{|\vec{R} + \frac{\vec{d}}{2}|^3} \right]$$

$d \ll R$



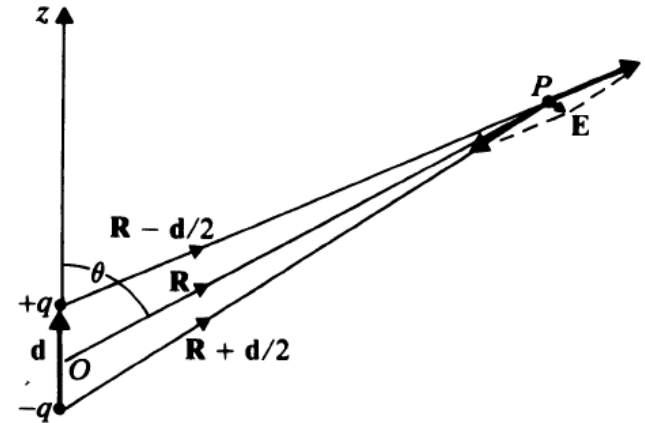
$$\vec{E} \approx \frac{q}{4\pi\epsilon_0 R^3} \frac{\vec{R} \times \vec{d}}{R} = \frac{1}{4\pi\epsilon_0} \frac{\vec{r} \times \nabla}{R^3}$$

注：电偶极矩的电场强度在数值上与距离三次方成反比（且不再各向同性，与角度 θ 有关），因此衰减更快。

◆ Electric dipole moment, p : $\vec{p} = q\vec{d}$



$$\vec{E} = \frac{1}{4\pi\epsilon_0 R^3} \left[\frac{\vec{R} \times \vec{p}}{R^2} \times \vec{R} - \frac{\vec{p}}{R^3} \right]$$



If p is along z axis, then $\vec{p} = \vec{a}_z p = p(\vec{a}_R \cos \theta - \vec{a}_\theta \sin \theta)$

◆ Electric field $\vec{E}$ of a electric dipole in spherical coordinate

$$\vec{E} = \frac{p}{4\pi\epsilon_0 R^3} (\vec{a}_R 2 \cos \theta + \vec{a}_\theta \sin \theta)$$

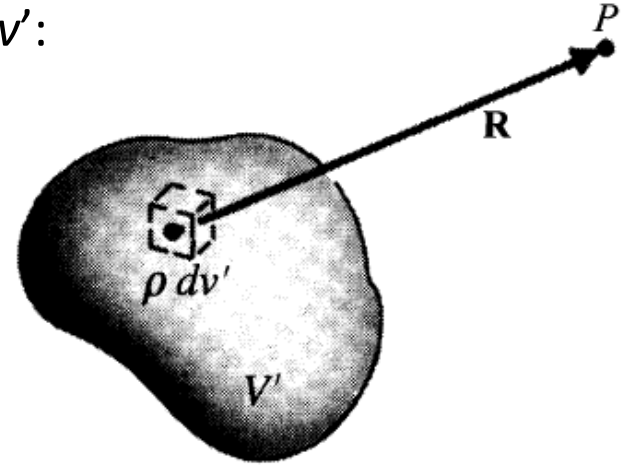


Cubic dependence on R

3.3.2 Electric field due to a continuous distribution of charge

◆ Electric field $\mathbf{E}$ of a differential volume charge $\rho dv'$:

$$\vec{E} = \vec{a}_R \frac{\rho dv'}{4\pi\epsilon_0 R^2} \quad (\vec{a}_R = \vec{R} / R)$$



◆ Total electric field $\mathbf{E}$:

$$(1) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \oint_{V'} \vec{a}_R \frac{\rho}{R^2} dv'$$

$$(2) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \oint_{S'} \vec{a}_R \frac{\rho_s}{R^2} ds'$$

$$(3) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \oint_{L'} \vec{a}_R \frac{\rho_l}{R^2} dl'$$

(ρ : volume charge density)

带撇的坐标表示源的位置，不带撇的坐标，表示我们要计算的场强的位置（点）。

(ρ_s : surface charge density)

(ρ_l : line charge density)

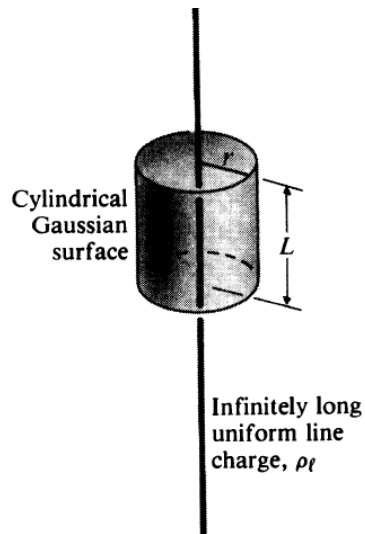
3.4 Gauss's Law and Applications

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

Suitable for symmetry condition with proper surface

用高斯定律很容易求出一些具有对称性的电荷分布的电场强度（如无限长的线电荷、无限宽的面电荷）

Example 1: Find $\vec{E}$ for an infinite uniform line charge



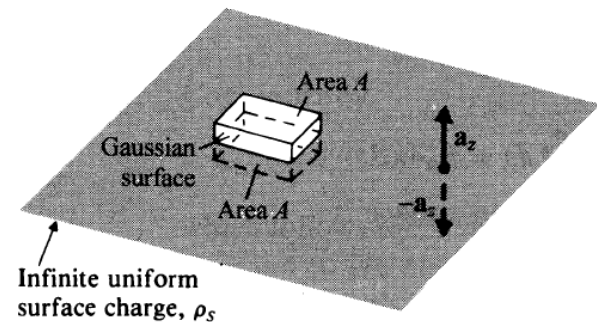
$$\oint \vec{E} \cdot d\vec{s} = \oint_0^L \oint_0^{2\pi} E_r r df dz$$

$$= 2\pi r L E_r$$

$$\Rightarrow 2\pi r L E_r = \frac{r_l L}{\epsilon_0}$$

$$\Rightarrow \vec{E} = \vec{a}_r E_r = \vec{a}_r \frac{r_l}{2\pi\epsilon_0 r}$$

Example 2: Find $\vec{E}$ for an infinite uniform surface charge



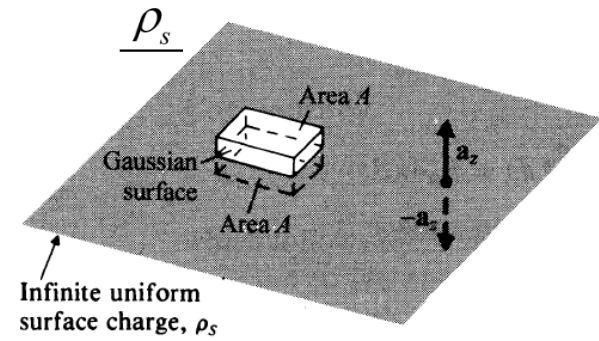
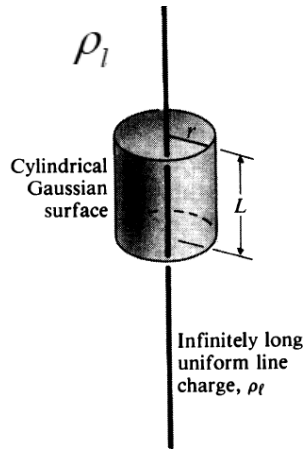
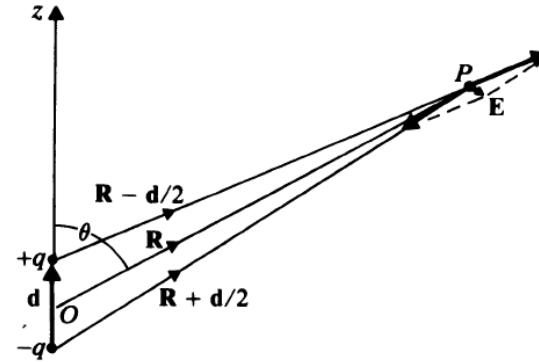
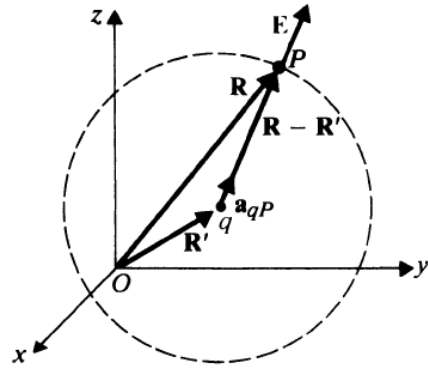
$$\vec{E} \cdot d\vec{s} = (\vec{a}_z E_z) \cdot (\vec{a}_z ds) = E_z ds$$

$$\Rightarrow \vec{E} = \vec{a}_z E_z = \vec{a}_z \frac{\rho_s}{2\epsilon_0}, (z > 0)$$

$$\vec{E} = -\vec{a}_z E_z = -\vec{a}_z \frac{\rho_s}{2\epsilon_0}, (z < 0)$$

Free space

 $\nabla \equiv$ $\nabla^2 =$ $\nabla V =$ $\nabla \cdot \mathbf{A} =$ $\nabla \times \mathbf{A} =$ $\nabla^2 V =$ $\nabla \cdot \vec{E} =$ $\nabla \times \vec{E} =$ $\oint_S \vec{E} \cdot d\vec{s} =$ $\oint_C \vec{E} \cdot d\vec{l} =$



$$\nabla \equiv \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

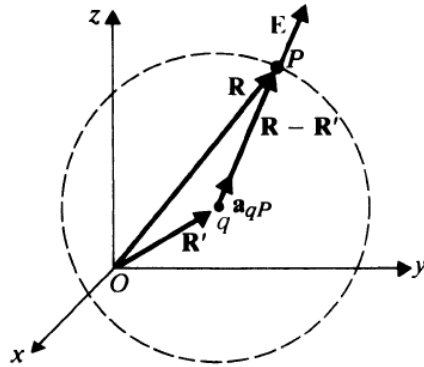
$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \mathbf{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \mathbf{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \mathbf{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

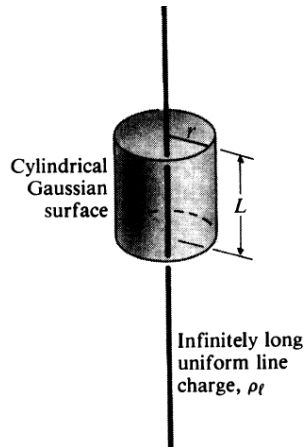
$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\varepsilon_0}$$

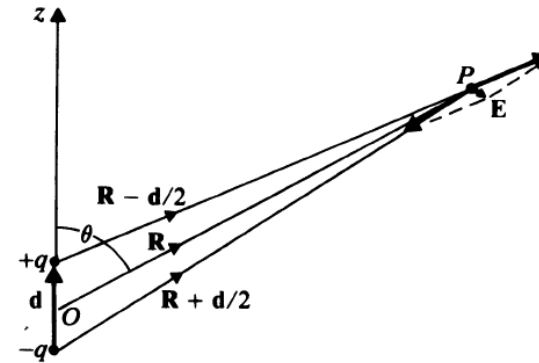
$$\nabla \times \vec{E} = 0 \qquad \oint_C \vec{E} \cdot d\vec{l} = 0$$



$$\vec{E}_p = \frac{q(\vec{R} - \vec{R}')}{4\pi\epsilon_0 |\vec{R} - \vec{R}'|^3}$$



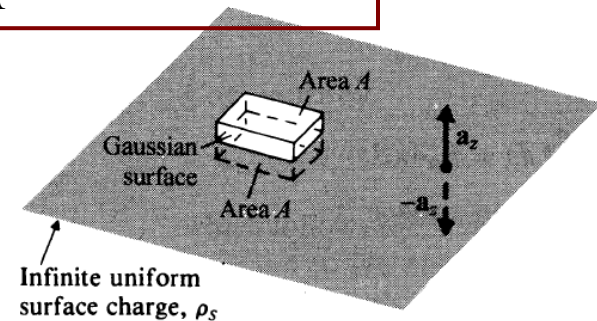
$$\vec{E} = \vec{a}_r E_r = \vec{a}_r \frac{\rho_l}{2\pi\epsilon_0 r}$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0 R^3} \left(\frac{\vec{R} \times \vec{p}}{R^2} \vec{R} - \vec{p} \right)$$

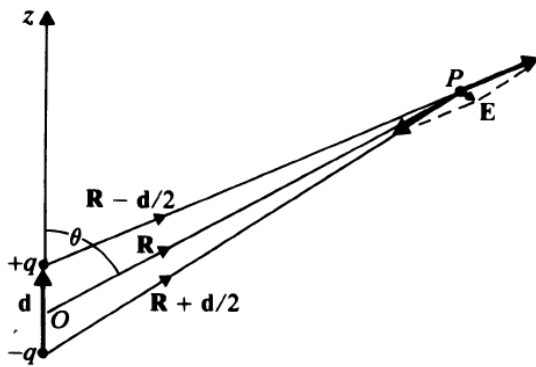
$$\vec{p} = q\vec{d}$$

$$\vec{E} = \frac{p}{4\pi\epsilon_0 R^3} (\vec{a}_R 2 \cos \theta + \vec{a}_\theta \sin \theta)$$



$$\vec{E} = \vec{a}_z E_z = \vec{a}_z \frac{\rho_s}{2\epsilon_0}, \quad (z > 0)$$

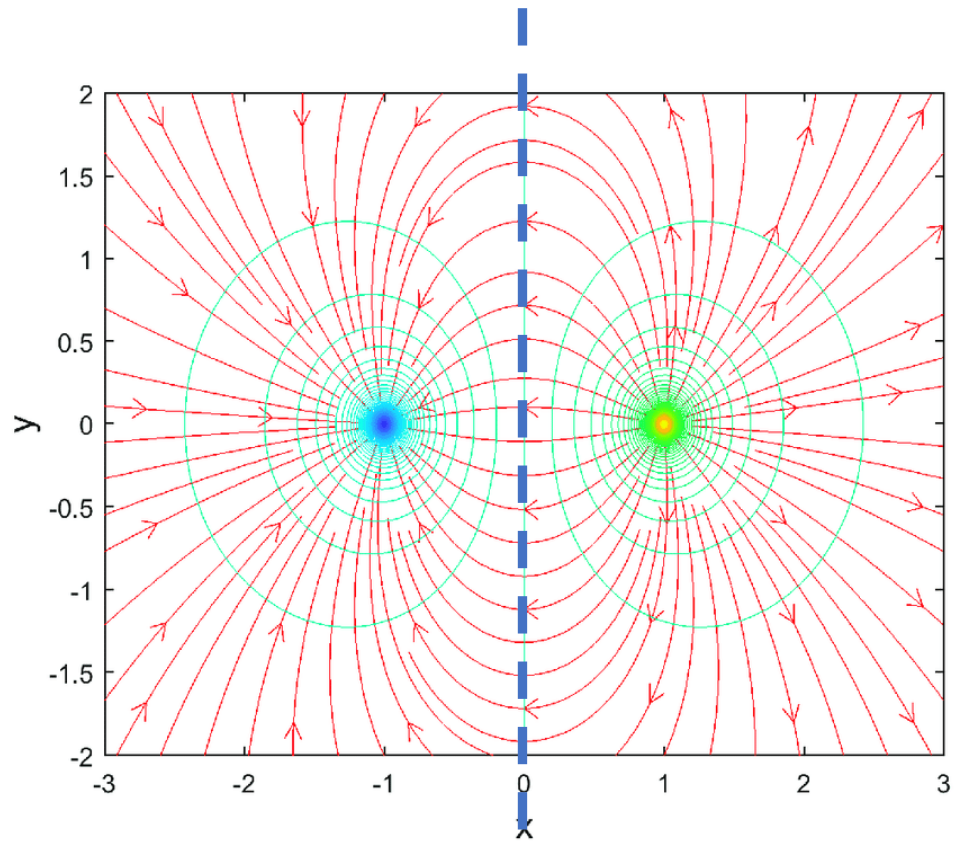
$$\vec{E} = -\vec{a}_z E_z = -\vec{a}_z \frac{\rho_s}{2\epsilon_0}, \quad (z < 0)$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0 R^3} \left(\frac{\vec{R} \times \vec{p}}{R^2} \times \vec{R} - \vec{p} \right)$$

$$\vec{E} = \frac{p}{4\pi\epsilon_0 R^3} (\vec{a}_R 2 \cos \theta + \vec{a}_\theta \sin \theta)$$

$$\vec{p} = q\vec{d}$$



3.5 Electric Potential

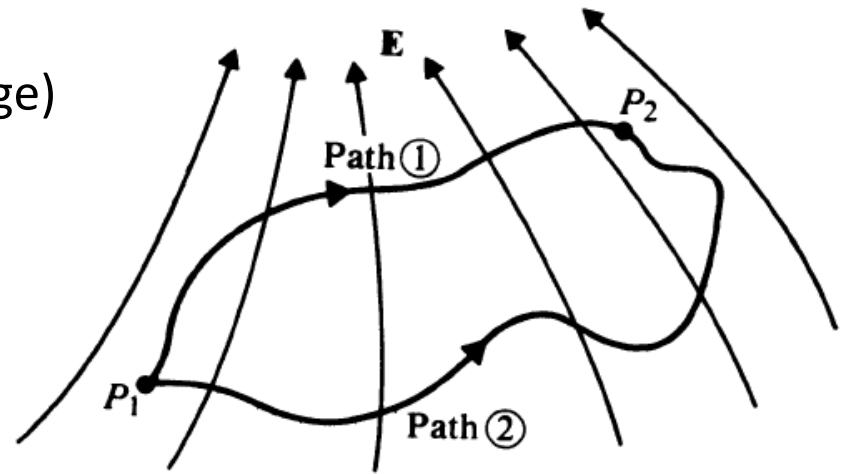
- ◆ A scalar electric potential V :

因为 $\vec{E}$ 的旋度为零，这意味着 $\vec{E}$ 必然可以表示为一个标量的梯度（因为梯度的旋度恒为零）

$$\boxed{\vec{E} = -\nabla V} \iff \begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \times (\nabla A) = 0 \end{cases}$$

- ◆ Potential difference (electrostatic voltage) between points P_2 and P_1 :

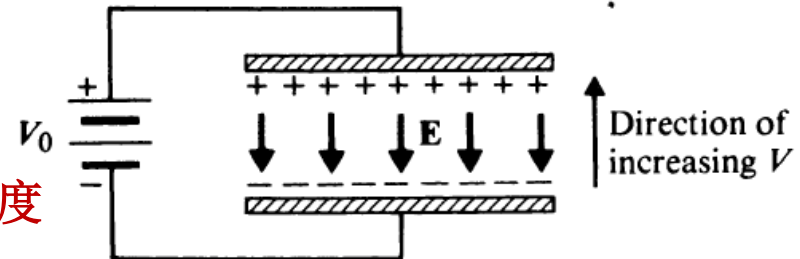
$$\boxed{V_2 - V_1 = - \int_{P_1}^{P_2} \vec{E} \cdot d\vec{l}}$$



In most cases, the zero-potential point is taken at infinity.

- ◆ Field lines are perpendicular to equipotential lines (surface) everywhere.

电力线垂直于等势线，因为 $\vec{E}$ 是势 V 的梯度（梯度是最大方向导数，是势下降最快的方向）。



3.5.1 Electric Potential due to a charge distribution

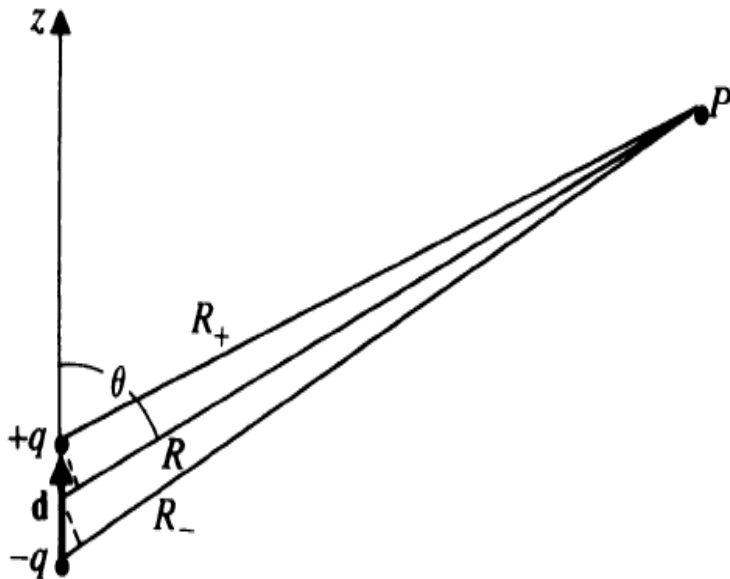
◆ Electric potential of a point at a distance R from a point charge q :

$$V = \frac{q}{4\pi\epsilon_0 R}$$



$$V = - \int_{\infty}^R (\vec{a}_R \frac{q}{4\pi\epsilon_0 R^2}) \cdot (\vec{a}_R dR)$$

◆ Electric potential of a point at a distance R from an electric dipole $\vec{p}$:



$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_+} - \frac{1}{R_-} \right).$$



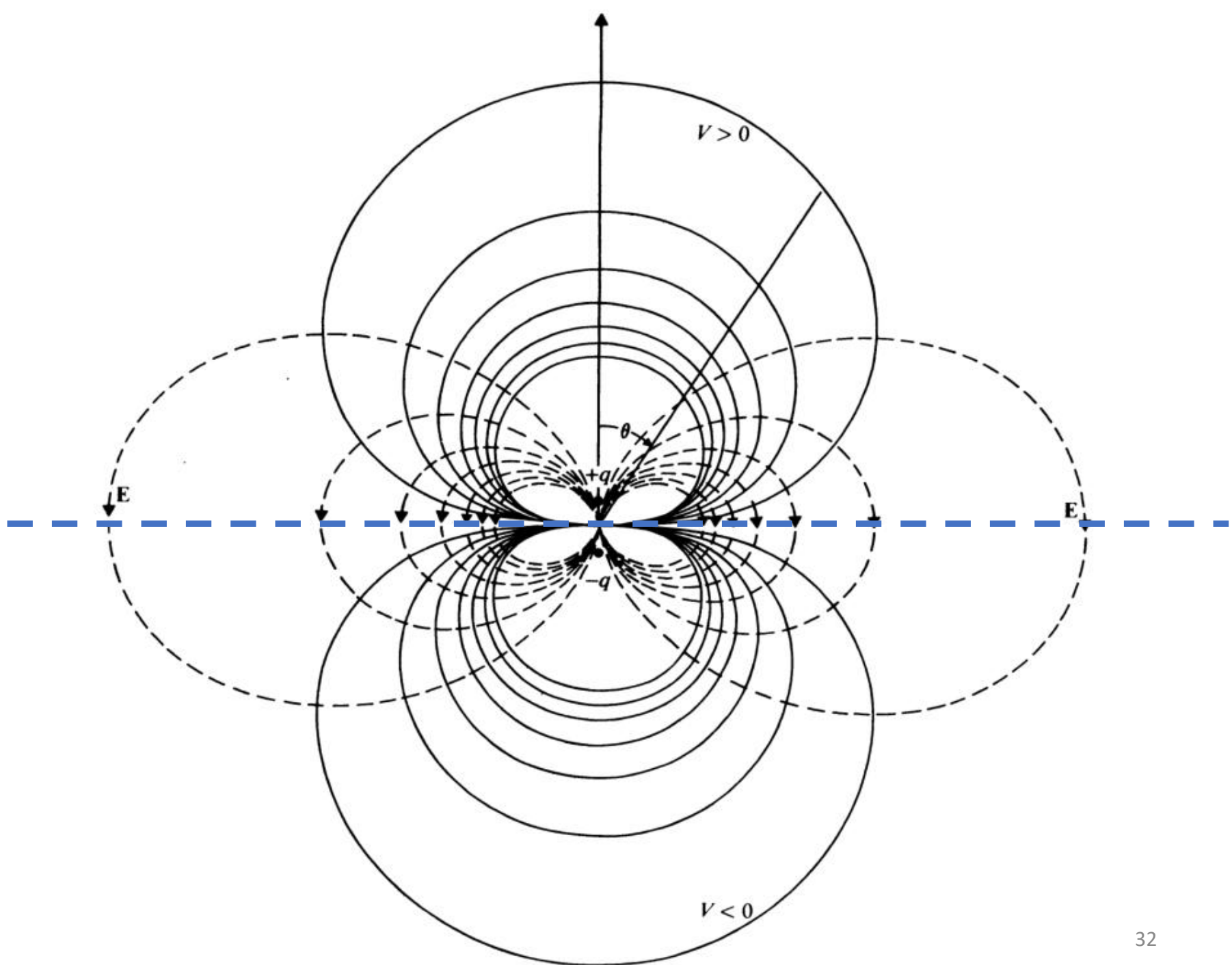
$d \ll R$

$$\frac{1}{R_+} \cong \left(R - \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 + \frac{d}{2R} \cos \theta \right)$$

$$\frac{1}{R_-} \cong \left(R + \frac{d}{2} \cos \theta \right)^{-1} \cong R^{-1} \left(1 - \frac{d}{2R} \cos \theta \right).$$

$$V = \frac{\vec{p} \cdot \vec{a}_R}{4\pi\epsilon_0 R^2} \quad (\text{V}),$$

$$\vec{p} = q\vec{d}$$



◆ Electric potential at $\mathbf{R}$ from a charge distribution:

(a) From n charges

$$V = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{q_k}{|\vec{R} - \vec{R}'_k|}$$

(b) From charges of volume density ρ

$$V = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{R}')}{|\vec{R} - \vec{R}'|} dv$$

P

Easier to determine $\vec{E}$ by taking the negative gradient, i.e.,

$$\vec{E} = -\nabla V$$

3.6 Conductors in Static Electric Field

◆ **Conductor, semiconductor, insulator (dielectric)**

◆ Inside a conductor, under static conditions we have

$$r = 0, \quad \vec{E} = 0$$

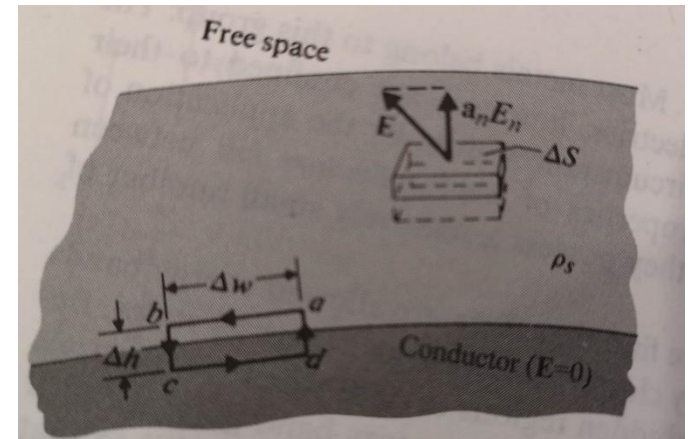
◆ **Boundary conditions** at a conductor/free-space interface:

Tangent component: (导体表面，切向电场必为零。如果不为零，就会驱动电荷，迅速重新排布，直到使得切向电场为零为止):

$$\oint_{\text{abcd}} \vec{E} \cdot d\vec{l} = E_t \Delta w = 0 \Rightarrow E_t = 0$$

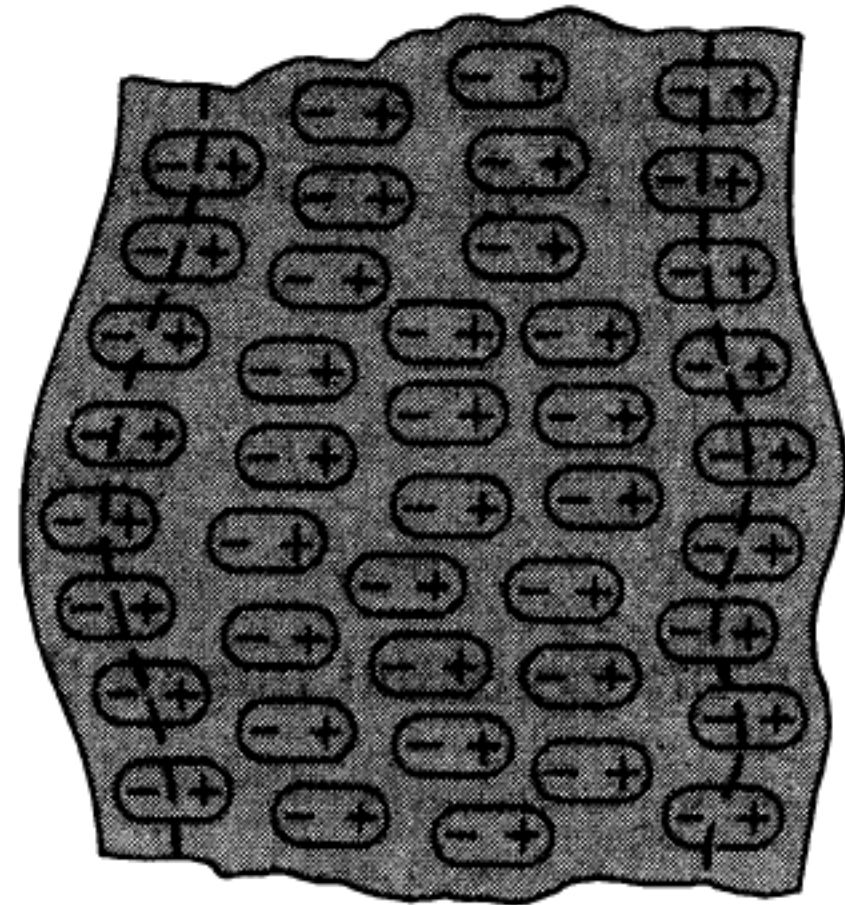
Normal component:

$$\oint_S \vec{E} \cdot d\vec{S} = E_n \Delta S = \frac{\rho_s \Delta S}{\epsilon_0} \Rightarrow E_n = \frac{\rho_s}{\epsilon_0}$$



For a neutral conductor placed in a static field, the induced electric field will cancel the external field both inside the conductor and tangent to its surface. (这就是静电屏蔽效应，导体是一个等势体，内部 $E=0$ 。只要导体还不是等势体，内部 E 不为零，就会驱动内部的自由电子气到表面，使得电荷在导体表面迅速重新排布，直到内部 E 被抵消为止，不再驱动内部自由电子，导体就成为了等势体。所以，导体总是一个等势体)

3.7 Dielectrics in Static Electric Field



Free charges vs bound charges 束缚电荷

Induced electric dipoles

在外电场作用下，均匀电介质（绝缘体）内部净电荷数为零，但在表面上显示有极化电荷 (polarized charge)（注：它们不是自由电荷）

◆ Polarization vectors $\mathbf{P}$ (极化强度矢量)
含义: 单位体积内的电偶极矩 $\mathbf{p}$

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{p}_k}{\Delta v} \quad (\text{C/m}^2),$$

$\mathbf{P}$ is the volume density of electric dipole moment $\mathbf{p}$. 电偶极矩

The dipole moment $d\mathbf{p}$ of an element volume dv' is $d\mathbf{p} = \mathbf{P}dv'$, which produces a potential:

$$dV = \frac{\mathbf{P} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} dv'.$$

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2} \quad (\text{V}),$$

The potential V due to the dielectric:

$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\mathbf{P} \cdot \mathbf{a}_R}{R^2} dv',$$

(R is the distance from the volume dv' to the fixed field point)

$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\mathbf{P} \cdot \mathbf{a}_R}{R^2} dv$$

$$\nabla' \left(\frac{1}{R} \right) = \frac{\mathbf{a}_R}{R^2}$$

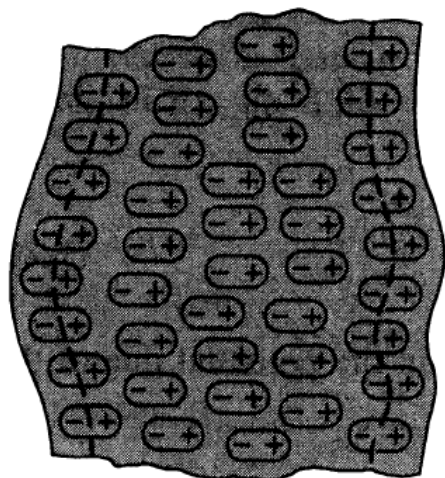
$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \mathbf{P} \cdot \nabla' \left(\frac{1}{R} \right) dv$$

$$\nabla' \cdot (f\mathbf{A}) = f\nabla' \cdot \mathbf{A} + \mathbf{A} \cdot \nabla' f$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\int_{V'} \nabla' \cdot \left(\frac{\mathbf{P}}{R} \right) dv' - \int_{V'} \frac{\nabla' \cdot \mathbf{P}}{R} dv' \right]$$

Divergence theorem

$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\mathbf{P} \cdot \mathbf{a}'_n}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{(-\nabla' \cdot \mathbf{P})}{R} dv'$$



External $\mathbf{E}$

微分表达形式
(点的形式)

表面极化电荷密度 (面电荷密度)
与 $\mathbf{P}$ (极化强度) 在数值上相等

Polarization charge densities or bound-charge densities

$$\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n$$

$$\rho_p = -\nabla \cdot \mathbf{P}$$

极化体电荷密度 (polarized
volume charge density)

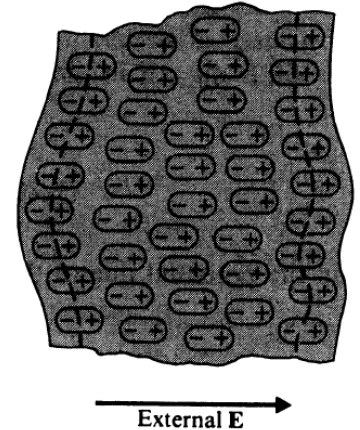
$$V = \frac{1}{4\pi\epsilon_0} \oint_{S'} \frac{\rho_{ps}}{R} ds' + \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_p}{R} dv'$$

3.8 Electric Flux Density D and Dielectric Constant

◆ Gauss's equation in the dielectric:

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} (\overset{\text{Free charges}}{\rho} + \overset{\text{Polarized charges}}{\rho_p}) \quad \text{or} \quad \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho.$$

$$\rho_p = -\nabla \cdot \mathbf{P}$$



◆ Electric flux density or electric displacement, $\mathbf{D}$:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

Gauss's equation

$$\nabla \cdot \mathbf{D} = \rho$$

(Differential form)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

Free charges (自由电荷)

(Integral form)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

The total outward flux of the electric displacement over any closed surface is equal the total free charge enclosed in the surface.

◆ Polarization for **linear**(线性) and **isotropic**(各项同性) system:

$$\begin{aligned} \mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P} \\ \mathbf{P} &= \epsilon_0 \chi_e \mathbf{E} \\ \mathbf{D} &= \epsilon_0 (1 + \chi_e) \mathbf{E} \\ &= \epsilon_0 \epsilon_r \mathbf{E} = \epsilon \mathbf{E} \end{aligned}$$


χ_e: Electric susceptibility 电极化率

Where **relative permittivity or dielectric constant** is defined by

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$$

Dielectric constant (Complex in general)

Dielectric constant也可以是张量（称为介电张量，用来描述各向异性材料）

Anisotropic medium

（各向异性介质）

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$



Biaxial medium

（双轴各向异性介质）

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$



Isotropic medium

（各向同性介质）

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_1 & 0 \\ 0 & 0 & \epsilon_1 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$



Uniaxial anisotropic medium

（单轴各向异性介质）

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_1 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

3.9 Boundary Conditions for Electrostatic Fields

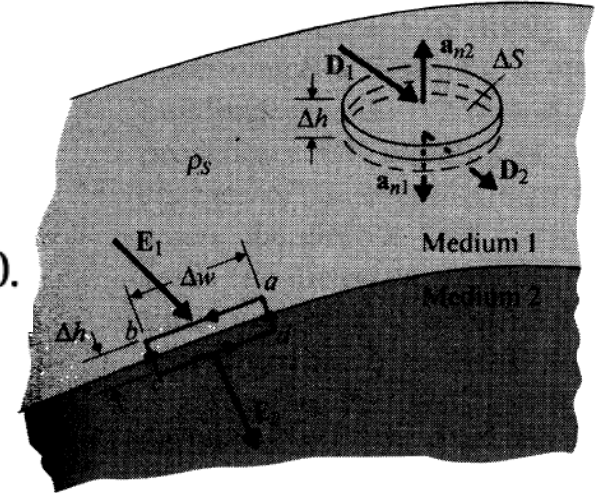
At the interface of two general media, we write the line integral of $\mathbf{E}$ around a closed path by

(1)

$$\oint_{\text{abcd}} \mathbf{E} \cdot d\boldsymbol{\ell} = \mathbf{E}_1 \cdot \Delta \mathbf{w} + \mathbf{E}_2 \cdot (-\Delta \mathbf{w}) = E_{1t} \Delta w - E_{2t} \Delta w = 0.$$

Therefore,

$$E_{1t} = E_{2t}$$



● *The tangential component of an $\mathbf{E}$ field is continuous.*

(2)

$$\begin{aligned} \oint_S \mathbf{D} \cdot d\mathbf{s} &= (\mathbf{D}_1 \cdot \mathbf{a}_{n2} + \mathbf{D}_2 \cdot \mathbf{a}_{n1}) \Delta S \\ &= \mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) \Delta S \\ &= \rho_s \Delta S, \end{aligned}$$

$$\longrightarrow D_{1n} - D_{2n} = \rho_s$$

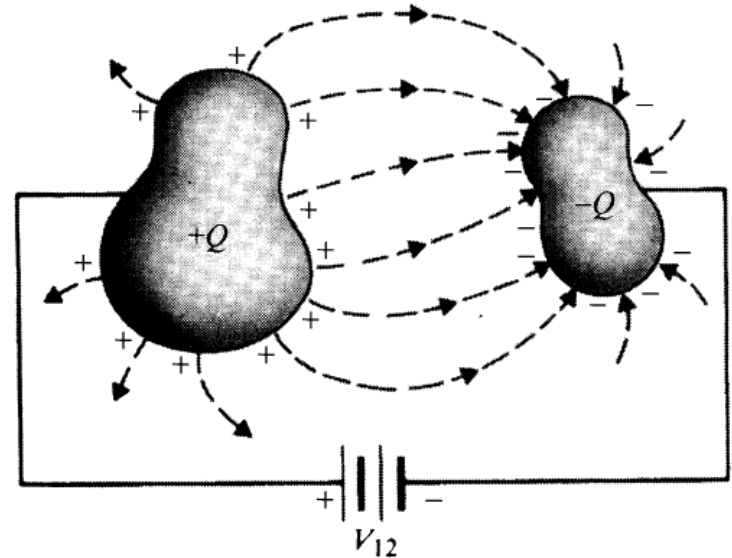
● *The normal component of $\mathbf{D}$ field is discontinuous and the discontinuity is equal to the surface charges.*

3.10 Capacitance and Capacitors

◆ Capacitance C :

$$C = \frac{Q}{V_{12}} \quad (\text{F}).$$

C is dependent on the geometry of the conductors and on the permittivity the surrounding medium.



3.11.1 Electrostatic Energy In Terms of Field Quantities

$$W_e = \frac{1}{2} \int_{V'} \mathbf{D} \cdot \mathbf{E} dv$$

$$W_e = \frac{1}{2} \int_{V'} \epsilon E^2 dv \quad (\text{J})$$

$$W_e = \frac{1}{2} \int_{V'} \frac{D^2}{\epsilon} dv \quad (\text{J}).$$

◆ Electrostatic energy density w_e

$$W_e = \int_{V'} w_e dv.$$

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} \quad (\text{J/m}^3)$$

$$w_e = \frac{1}{2} \epsilon E^2 \quad (\text{J/m}^3)$$

$$w_e = \frac{D^2}{2\epsilon} \quad (\text{J/m}^3).$$

Chapter 4: Solutions of Electrostatic Problems

4-1 Introduction

4-2 Poisson's and Laplace's Equations

4-3 Uniqueness of Electrostatic Solutions

4-4 Methods of Images

4-5 Boundary-Value Problems in Cartesian Coordinates

4-6 Boundary-Value Problems in Cylindrical Coordinates

4-7 Boundary-Value Problems in Spherical Coordinates

4.2 Poisson's and Laplace's Equations

Maxwell Equation:

$$\nabla \cdot \vec{D} = \rho \quad \text{free charges}$$

$$\nabla \times \vec{E} = 0$$

$$\vec{E} = -\nabla V$$

In a linear and isotropic medium: $\vec{D} = \epsilon \vec{E}$

$$\nabla \cdot \epsilon \vec{E} = \rho$$

$$\nabla \cdot (\epsilon \nabla V) = -\rho$$

For a homogeneous medium:

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

Poisson's equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$$

$$\nabla^2 V = \nabla \cdot \nabla V = \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right) \cdot \left(\vec{a}_x \frac{\partial V}{\partial x} + \vec{a}_y \frac{\partial V}{\partial y} + \vec{a}_z \frac{\partial V}{\partial z} \right)$$

$$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

4.2 Poisson's and Laplace's Equations

Cylindrical coordinates:

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \Phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical coordinates:

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \Phi^2}$$

no free charge Laplace's equation

$$\nabla^2 V = 0$$

electric field

$$\vec{E} = -\nabla V$$

**charge distribution on the
conductor surfaces**

$$\rho_s = \epsilon \vec{E}_n$$

Determine the field both inside and outside a spherical cloud of electrons with a uniform volume charge density $\rho = -\rho_0$ (where ρ_0 is a positive quantity) for $0 \leq R \leq b$ and $\rho = 0$ for $R > b$ by solving Poisson's and Laplace's equation for V .

(1) Inside the cloud $0 \leq R \leq b$ $\rho = -\rho_0$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

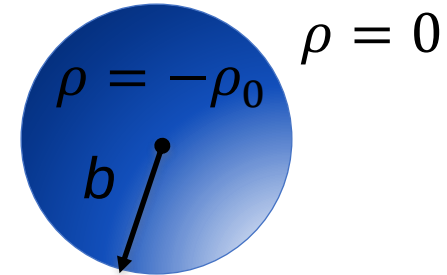
$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} \qquad \frac{d}{dR} \left(R^2 \frac{dV_i}{dR} \right) = \frac{\rho_0}{\epsilon_0} R^2$$

$$\frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R + \frac{C_1}{R^2}.$$

The electric field intensity inside

$$\vec{E}_i = -\nabla V_i = -\vec{a}_R \left(\frac{dV_i}{dR} \right)$$

$$\vec{E}_i = -\vec{a}_R \left(\frac{\rho_0}{3\epsilon_0} \right) R$$



$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \Phi^2}$$

Determine the field both inside and outside a spherical cloud of electrons with a uniform volume charge density $\rho = -\rho_0$ (where ρ_0 is a positive quantity) for $0 \leq R \leq b$ and $\rho = 0$ for $R > b$ by solving Poisson's and Laplace's equation for V .

(2) Outside the cloud $b \leq R$ $\rho = 0$

Laplace's equation $\nabla^2 V = 0$

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dV_o}{dR} \right) = 0$$

$$\frac{dV_o}{dR} = \frac{C_2}{R^2}$$

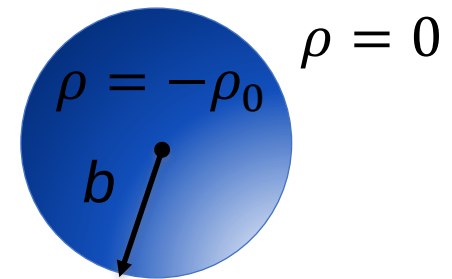
The electric field $\vec{E}_0 = -\nabla V_0 = -\vec{a}_R \left(\frac{dV_0}{dR} \right) = -\vec{a}_R \left(\frac{C_2}{R^2} \right)$

The electric field continuity at $R=b$ $\frac{C_2}{b^2} = \frac{\rho_0}{3\epsilon_0} b$

$$\vec{E}_0 = -\vec{a}_R \frac{\rho_0 b^3}{3\epsilon_0 R^2}$$

$$Q = -\rho_0 \frac{4\pi}{3} b^3$$

$$\vec{E}_0 = \vec{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$$



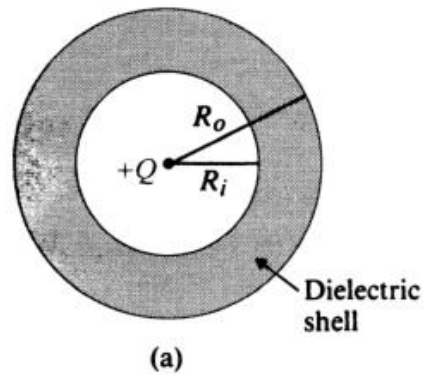
inside $\frac{dV_i}{dR} = \frac{\rho_0}{3\epsilon_0} R + \frac{C_1}{R^2}.$

4.3 Uniqueness of Electrostatic Solutions

Uniqueness theorem

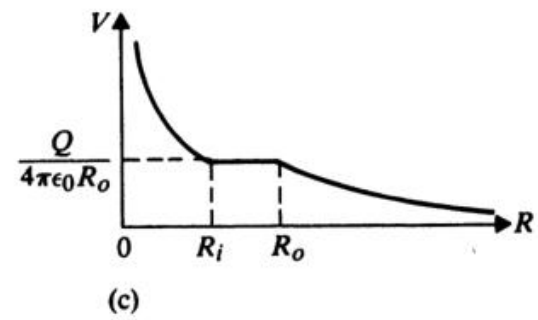
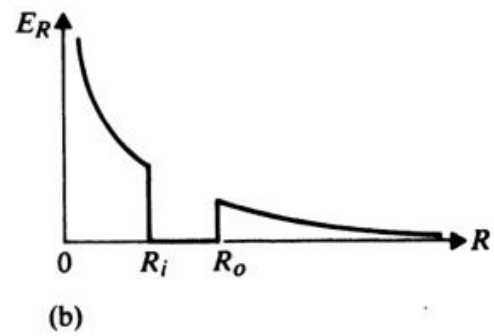
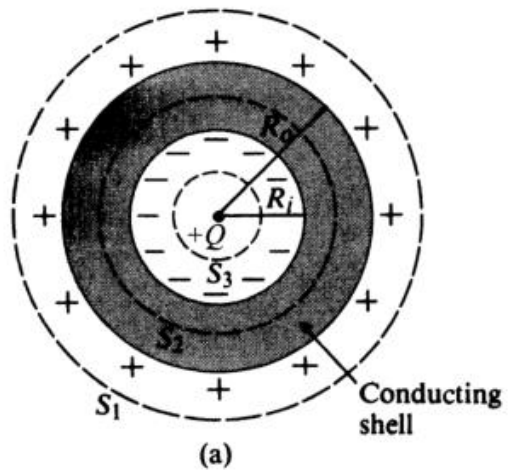
A solution of Poisson's equation (of which Laplace's equation is a special case) that satisfies the given boundary conditions is *a unique solution*.

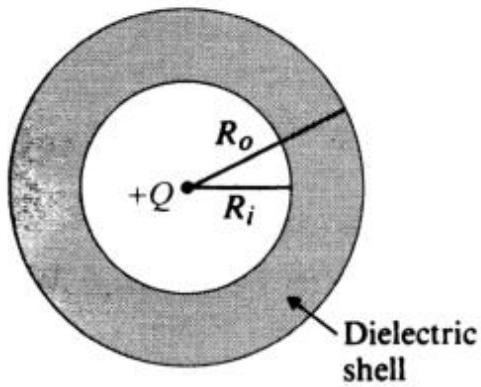
A solution of an electrostatic problem satisfying its boundary conditions is the only possible solution, *irrespective of the method by which the solution is obtained*.



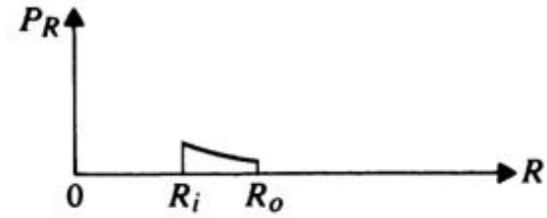
EXAMPLE 3–11 A positive point charge Q is at the center of a spherical conducting shell of an inner radius R_i and an outer radius R_o . Determine $\mathbf{E}$ and V as functions of the radial distance R .

EXAMPLE 3–12 A positive point charge Q is at the center of a spherical dielectric shell of an inner radius R_i and an outer radius R_o . The dielectric constant of the shell is ϵ_r . Determine $\mathbf{E}$, V , $\mathbf{D}$, and $\mathbf{P}$ as functions of the radial distance R .

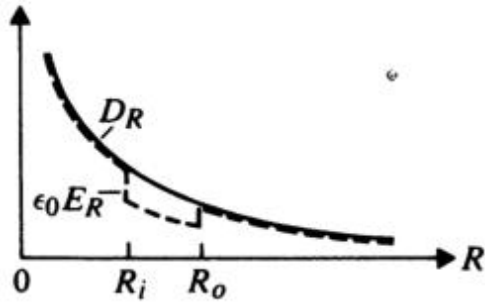




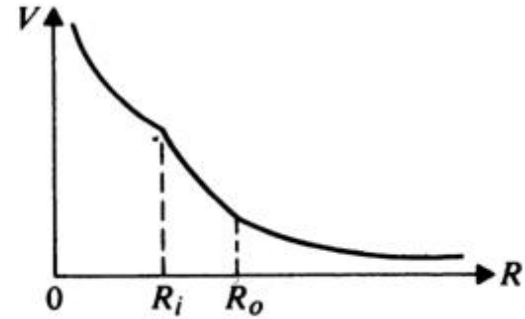
(a)



(c)



(b)



(d)

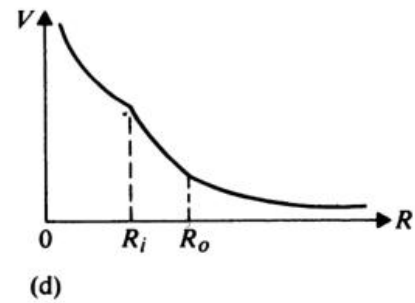
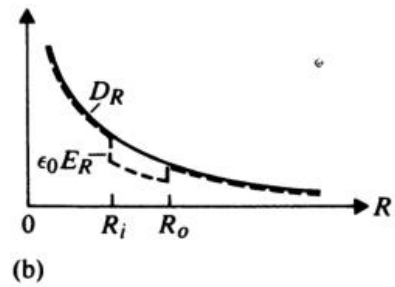
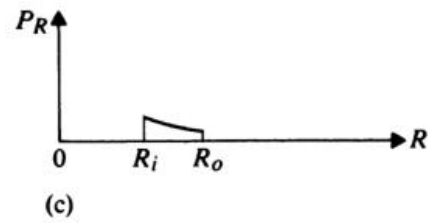
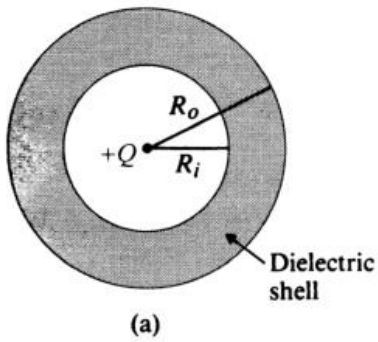
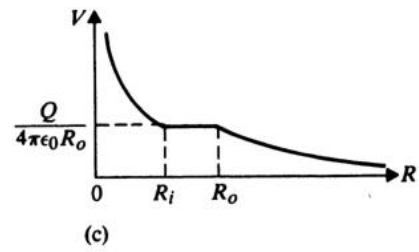
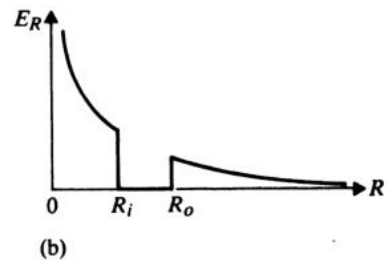
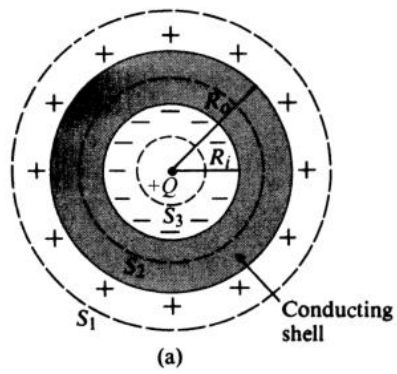
$$\begin{aligned}\rho_{ps}|_{R=R_i} &= \mathbf{P} \cdot (-\mathbf{a}_R)|_{R=R_i} = -P_{R2}|_{R=R_i} \\ &= -\left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_i^2}\end{aligned}$$

on the inner shell surface,

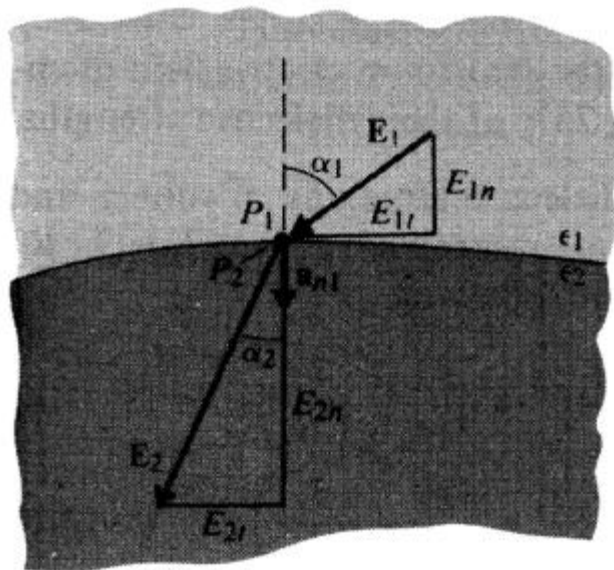
$$\begin{aligned}\rho_{ps}|_{R=R_o} &= \mathbf{P} \cdot \mathbf{a}_R|_{R=R_o} = P_{R2}|_{R=R_o} \\ &= \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R_o^2}\end{aligned}$$

on the outer shell surface, and

$$\begin{aligned}\rho_p &= -\nabla \cdot \mathbf{P} \\ &= -\frac{1}{R^2} \frac{\partial}{\partial R} (R^2 P_{R2}) = 0.\end{aligned}$$



EXAMPLE 3-15 Two dielectric media with permittivities ϵ_1 and ϵ_2 are separated by a charge-free boundary as shown in Fig. 3-25. The electric field intensity in medium 1 at the point P_1 has a magnitude E_1 and makes an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at point P_2 in medium 2.



EXAMPLE 3-16 When a coaxial cable is used to carry electric power, the radius of the inner conductor is determined by the load current, and the overall size by the voltage and the type of insulating material used. Assume that the radius of the inner conductor is 0.4 (cm) and that concentric layers of rubber ($\epsilon_{rr} = 3.2$) and polystyrene ($\epsilon_{rp} = 2.6$) are used as insulating materials. Design a cable that is to work at a voltage rating of 20 (kV). In order to avoid breakdown due to voltage surges caused by lightning and other abnormal external conditions, the maximum electric field intensities in the insulating materials are not to exceed 25% of their dielectric strengths.

